Day-5

Java Method Syntax:

1.What is the syntax of a method in Java, and what are its main components?

```
accessModifier returnType method_name(parameter)
{
// method body }`
```

- accessModifier --> public,private,protected,default
- returnType -->void,int,double.String etc
- method_name --> identifier
- parameters -->input value
- method body ->logic

2.What is the difference between instance methods and static methods in Java?

Instance method /non-static

- it belongs to object
- needs obj to call

ex-

```
import java.util.*;

public class Main {

  static  class Test{
```

```
    public  void display(){

        System.out.println("just print the instance method");

    }

}

    public static void main(String[] args) {

    Test t1= new Test();

    t1.display();

    }

}
```

static Method

- it belong to class
- there is no need of object

```
import java.util.*;

public class Main {
```

```
class Test{

    static void display(){

        System.out.println("just print Static method");

    }

}

    public static void main(String[] args) {

    Test.display();

    }

}
```

3.What is the difference between a method declaration and a method definition?
method Declaration:

- only method signature means no method body is required
- ex - public void greet();

method defination

- there is a full method body with implimentation
- ex- public void greet(){ sysout("hi brother"); }

4.Can a Java method have multiple return statements? Explain with an example.
yes , but 1 at a time
ex-

```
public int show(int num){
if(num>0){
return 1 ;
}
else if(num<0){
return -1
}
else (num = 0){
return 0;
}
}
```

5.What are access modifiers in Java methods, and how do they affect visibility?
access modifiers are used to control the visibility : means who can access

- public -> access for everywhere
- protected -> same package +sub class
- private -> with same class only
- default(no keyword) -> same package

Programs :

1. Write a Java method with no parameters that returns an integer value.
2. Write a static method that takes two numbers as parameters and returns their sum.
3. Write a public method that returns a string message.
4. Write a method that prints a message to the console but does not return any value.
5. Write a method that accepts parameters, performs a calculation, and returns the result.

Non-Parameterized Methods:
Interview Questions

1. Explain the purpose of non-parameterized methods in Java. When would you prefer using them?
   - a non-parameterized method does not take input
   - it is used when we need to reuse a common task
   - there is no needed of input

- ex- `void reuse(){ Sysout(" i am using non parameterized method here");}`

2. Can a non-parameterized method return different data types? Demonstrate with examples.

yes , it can return different data type like ex- `int greet(){ return "hi"; }`

3. Can a non-parameterized method access instance variables and static variables? Explain.

   yes ,

```
import java.util.*;

public class Main {

  int x = 10;           // instance variable

  static int y = 20;    // static variable

  void show() {

    System.out.println(x); // instance

    System.out.println(y); // static

  }

  public static void main(String[] args) {

  Main d =new Main();

  d.show();
```

```
    }

}
```

4. Can a non-parameterized method call another method that has parameters? Explain with an example.
   --- yes, but in parameterized just pass the value
5. What happens if you call a non-parameterized instance method before creating an object?
   👉 ❌ Not allowed

Programs:

1. Write a non-parameterized method that returns your name.
2. Write a non-parameterized method that returns the current year.
3. Write a non-parameterized method that calculates and returns the square of 5.
4. Write a non-parameterized method that prints a welcome message.
5. Write a non-parameterized method that returns a greeting message based on a predefined variable.

Methods with Parameters

1. What are parameters in a Java method, and why are they used?
2. What is the difference between formal parameters and actual parameters in Java?
3. Is Java pass-by-value or pass-by-reference? Explain with an example.
4. Can you pass an array or object as a parameter to a method in Java? Demonstrate with an example.
5. Can a method have multiple parameters? Explain with an example.

🔹 1) What are parameters? Why used?

👉 **Parameters = inputs to a method**

📌 Used to:

- Pass data
- Make method flexible

✅ Example:

```
void greet(String name) {
System.out.println("Hello " + name);
}
```

👉 `name` is a parameter

> 💡 Without parameters → same output every time
> With parameters → customizable output

🔹 2) Formal vs Actual Parameters

👉 **Formal Parameter** → in method definition
👉 **Actual Parameter** → value passed while calling

✅ Example:

```
void greet(String name) { // formal parameter
System.out.println(name);
}

greet("Aman"); // actual parameter
```

🔹 3) Pass-by-value or reference?

👉 Java is **always pass-by-value** ✅

📌 Meaning:

- Copy of value is passed
- Original doesn't change

- Example (primitive):

```
void change(int x) {
x = 50;
}

int a = 10;
change(a);

System.out.println(a); // 10 (no change)
```

- Example (object):

```
class Demo {
int x = 10;
}

void change(Demo d) {
d.x = 50;
}
```

👉 Object reference is copied → but object can change

🔹 4) Can we pass array/object?

👉 Yes ✅

✅ Array Example:

```
void printArray(int[] arr) {
System.out.println(arr[0]);
}
```

✅ Object Example:

```
class Student {
String name;
}

void printName(Student s) {
System.out.println(s.name);
}
```

🔹 5) Multiple Parameters

👉 Yes ✅
A method can take many inputs

✅ Example:

```
int add(int a, int b) {
return a + b;
}
```

👉 Call:

```
add(5, 3);
```

🎯 Final Memory Trick

> Parameters = inputs
> Formal = definition
> Actual = values
> Java = pass-by-value
> Methods can take multiple inputs

Programs

1. Write a method that accepts two integers and returns their sum.
2. Write a method that accepts a string parameter and returns its length.
3. Write a method that accepts an array of integers and returns the maximum element.
4. Write a method that accepts two numbers and returns the larger one.
5. Write a method that accepts multiple parameters, performs a calculation, and returns the result.

Method Returns Void

1. What does the void return type signify in a Java method?
2. Can a void method include a return statement? Explain with an example.
3. When should you use a void method instead of a method that returns a value?
4. Can a void method call another method that returns a value? Explain how.
5. What are the advantages and limitations of using void methods in Java?

🔹 1) What does `void` mean?

👉 `void` means **method returns nothing**

✅ Example:

```
void show() {
System.out.println("Hello");
}
```

📌 It only **does work**, gives no result back

🔹 2) Can void method have `return`?

👉 Yes ✅ but **without value**

✅ Example:

```
void check(int x) {
if (x < 0) {
return; // just stops method
}
System.out.println(x);
}
```

📌 `return;` is used to **exit method early**

🔹 3) When to use void method?

👉 Use when:

- You **don't need result**
- Just want to **perform action**

📌 Examples:

- Printing data
- Logging
- Display message

🔹 4) Can void method call return method?

👉 Yes ✅

✅ Example:

```
int add(int a, int b) {
return a + b;
}

void showSum() {
int result = add(2, 3); // calling return method
System.out.println(result);
}
```

📌 Void method can **use returned value internally**

🔹 5) Advantages & Limitations

✅ Advantages:

- Simple to use
- Good for actions (print, display)
- No need to handle return value

❌ Limitations:

- Cannot store result
- Cannot reuse output elsewher

Programs

1. Write a void method that prints "Hello, World!" to the console.
2. Write a void method that accepts a name as a parameter and prints a personalized greeting message.

3. Write a void method that prints numbers from 1 to 10 using a loop.
4. Write a void method that accepts an array of integers and prints all its elements.
5. Write a void method that performs a calculation but does not return the result.

Method Returns Value

1. What does it mean when a method returns a value in Java? Explain with an example.
2. How is the return type of a method declared, and why must it match the returned value?
3. Can a method return different types of values based on conditions? Explain with example.
4. What happens if the return statement in a method is missing or mismatched with the declared return type?
5. What are covariant return types in Java, and how are they used in method overriding?

Programs

1. Write a method that returns the factorial of a given number.
2. Write a method that checks if a number is prime and returns a boolean value.

3. Write a method that takes a string and returns it in reverse order.
4. Write a method that returns the area of a circle given its radius.
5. Write a method that returns the average of two numbers.

Factory Method – Return Type in Java

Interview Questions:

1. What is a Factory Method in Java, and how does it use return types to create objects?
2. What types of values can a method return in Java, and how are they used in factory methods?
3. What happens if a non-void method does not return a value? Explain with an example.
4. What is the difference between returning a primitive type and returning an object in Java?
5. What are covariant return types in Java, and how are they applied in factory method implementations?
6. How do return types enable polymorphism in Java, particularly in the context of factory methods?

Programs

Core Java Interview Questions

1. Write a factory method that returns an object based on the input type (e.g., Car, Bike).
2. Write a factory method that creates and returns different shapes (Circle, Rectangle) based on user input.
3. Write a factory method that returns an instance of a subclass depending on a condition.
4. Write a program demonstrating covariant return types using a base class and a subclass.
5. Write a factory method that returns null when no matching object type is found.
   Code Snippet Interview Questions

6. Basic Method Syntax
   Question:
   public class Test
   {
   public static void main(String[] args)
   {
   greet();
   }
   public static void greet() {
   System.out.println("Hello from Java!");
   }
   }
   Q: What is the output of this program?
   Q: Why is the greet() method declared as static?

7. Zero Parameter Method

Question:

```
public class Example
{
public static void main(String[] args)
{
displayMessage();
}
public static void displayMessage()
{
System.out.println("This is a zero-parameter method.");
}
```

Core Java Interview Questions

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}

Q: What will happen if you remove () from displayMessage while calling it?

Q: Can you overload the displayMessage() method?

3. Methods with Parameters

Question:

```
public class Demo
{
public static void main(String[] args)
{
int result = multiply(5, 3);
```

```
System.out.println(result);
}
public static int multiply(int a, int b)
{
return a b;
}
}
```

Q: What will be printed?

Q: What happens if you change the method's return type to void but still use return a b;?

4. Method Returns Void

Question:

```
public class VoidExample
{
public static void main(String[] args)
{
printSum(10, 5);
}
public static void printSum(int x, int y)
{
int sum = x + y;
System.out.println("Sum = " + sum);
```



```
}
}
```

Q: What happens if you try to store the result like int s = printSum(10, 5);?

5.Method Returns Value

Question:

```
public class ReturnExample
{
public static void main(String[] args)
{
int value = getSquare(4);
System.out.println("Square = " + value);
}
public static int getSquare(int num)
{
return num * num;
}
}
```

Q: What will be the output?

Q: What happens if you forget to include the return statement?

6. Method Overloading Example

   Question:

```
public class OverloadTest
{
public static void main(String[] args)
{
System.out.println(add(5, 10));
```

```
System.out.println(add(2.5, 3.5));
}
public static int add(int a, int b)
{
return a + b;
```